

Cognifyz Technologies

Data Analysis Internship

Restaurant Dataset Analysis

A comprehensive analysis covering Level 1, Level 2, and Level 3 internship tasks.

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B.E. Artificial Intelligence and Machine Learning

Tools: Python, Pandas, NumPy, Matplotlib, Plotly, NLTK, Jupyter Notebook

1. Project Overview

This project analyzes a restaurant dataset containing 9,551 records and 21 attributes. The analysis covers cuisine popularity, city distribution, price ranges, online delivery, ratings, cuisine combinations, geographic distribution, restaurant chains, votes, and the relationship between price range and restaurant services.

2. Dataset Summary

Metric	Value
Rows	9,551
Columns	21
Missing cuisine values	9
Average aggregate rating	2.6664
Average votes	156.91
Maximum votes	10,934

3. Level 1 — Basic Analysis

Task 1 — Top Cuisines

North Indian was the most common cuisine (3,960 restaurants; 41.50%), followed by Chinese (2,735; 28.66%) and Fast Food (1,986; 20.81%).

Task 2 — City Analysis

New Delhi had the highest number of restaurants with 5,473. The highest average rating in the city-level result was Inner City at 4.90.

Task 3 — Price Range Distribution

Price range 1 contained 4,444 restaurants (46.53%), followed by range 2 with 3,113 (32.59%), range 3 with 1,408 (14.74%), and range 4 with 586 (6.14%).

Task 4 — Online Delivery

2,451 restaurants (25.66%) offered online delivery, while 7,100 (74.34%) did not. The average rating was 3.2488 for restaurants with online delivery versus 2.4653 for those without.

4. Level 2 — Intermediate Analysis

Task 1 — Restaurant Ratings

The 3–4 rating range contained the most restaurants (4,388). The average number of votes was 156.91.

Task 2 — Cuisine Combination

The most common cuisine combination in the processed result was North Indian (936), followed by North Indian, Chinese (511). Within the analyzed top-20 summary, the correlation between restaurant count and average rating was -0.169, indicating a weak negative relationship.

Task 3 — Geographic Analysis

Latitude and longitude were available without missing values. The geographic analysis shows a strong concentration of restaurants in a small number of cities, particularly New Delhi (5,473), Gurgaon (1,118), and Noida (1,080). An interactive Plotly map was also generated.

Task 4 — Restaurant Chains

734 restaurant names appeared more than once and were treated as chains for this analysis. Cafe Coffee Day had 83 outlets, Domino's Pizza 79, and Subway 63 in the top-outlet result.

5. Level 3 — Advanced Analysis

Task 1 — Restaurant Reviews

The provided dataset does not contain free-form customer review text. The available 'Rating text' field is categorical (for example, Average, Good, Very Good, Excellent, Poor, and Not rated). Therefore, positive/negative keyword extraction and review-length analysis were not claimed.

Task 2 — Votes Analysis

Toit had the highest number of votes at 10,934. The correlation between Votes and Aggregate rating was 0.3137, interpreted in the notebook as a moderate positive correlation.

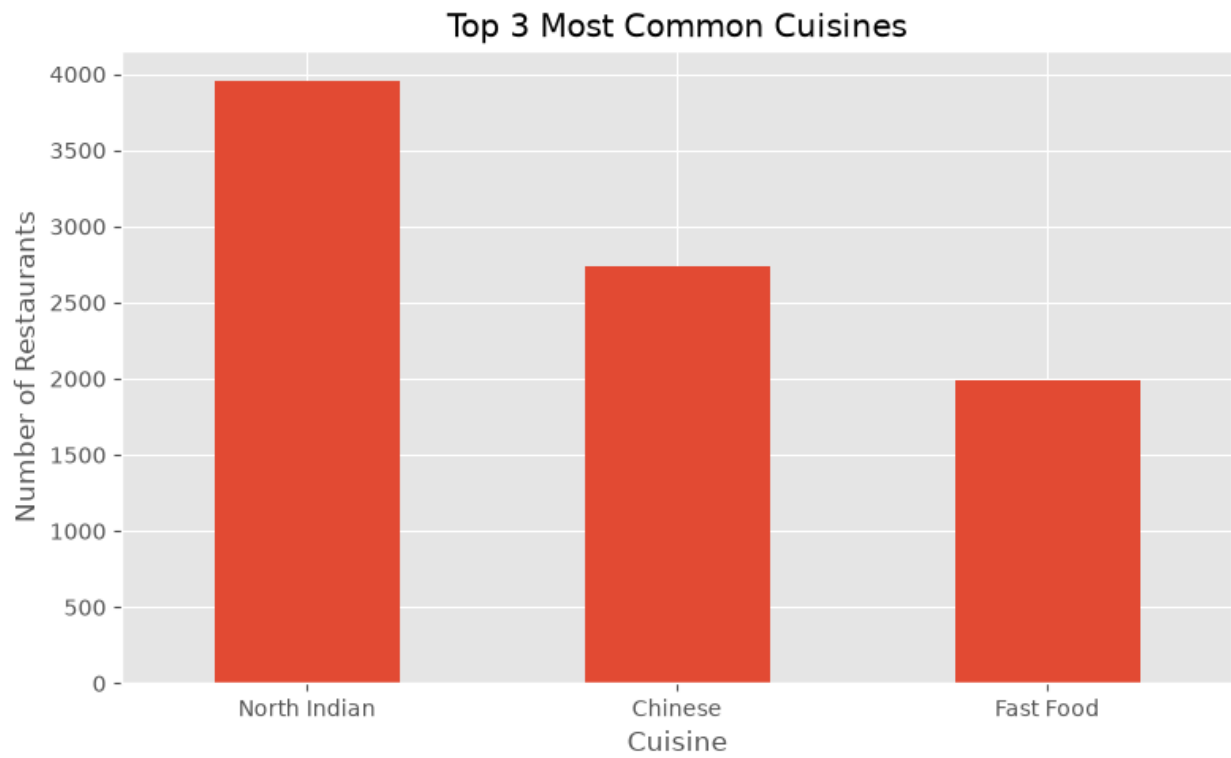
Task 3 — Price Range vs Services

Online-delivery availability was 15.77% for price range 1, 41.31% for range 2, 29.19% for range 3, and 9.04% for range 4. Table-booking availability increased from 0.02% in range 1 to 46.76% in range 4. Thus, higher price range was strongly associated with table-booking availability in this dataset, while online delivery did not increase monotonically with price range.

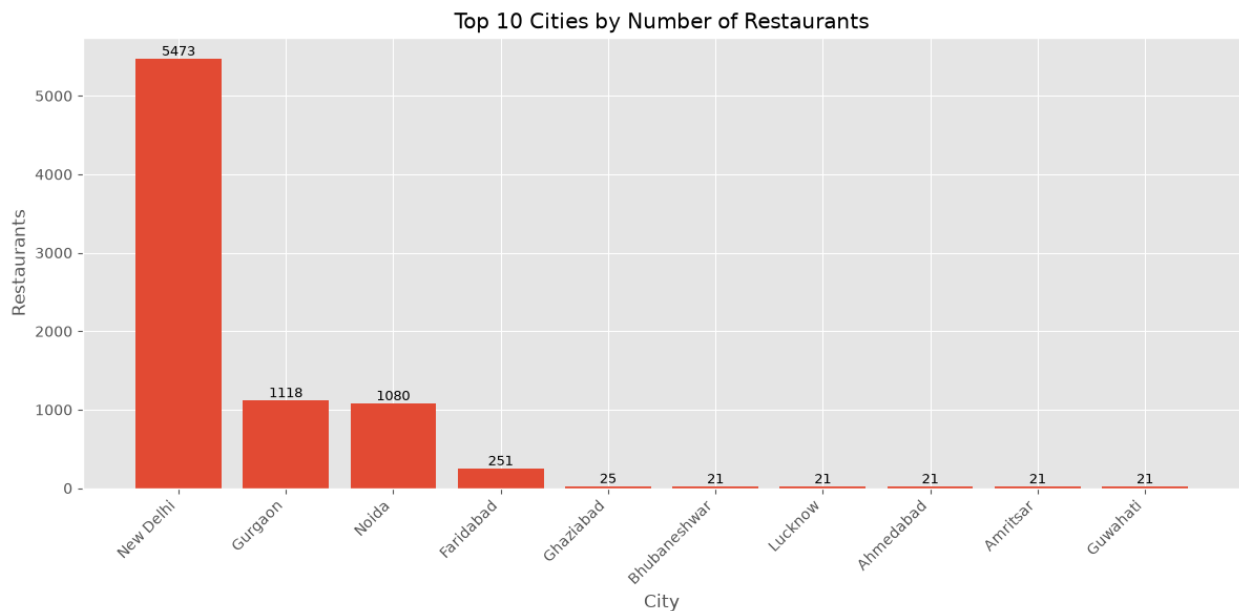
6. Selected Visualizations

The repository includes selected high-value visualizations extracted from the completed notebooks, plus an interactive geographic map.

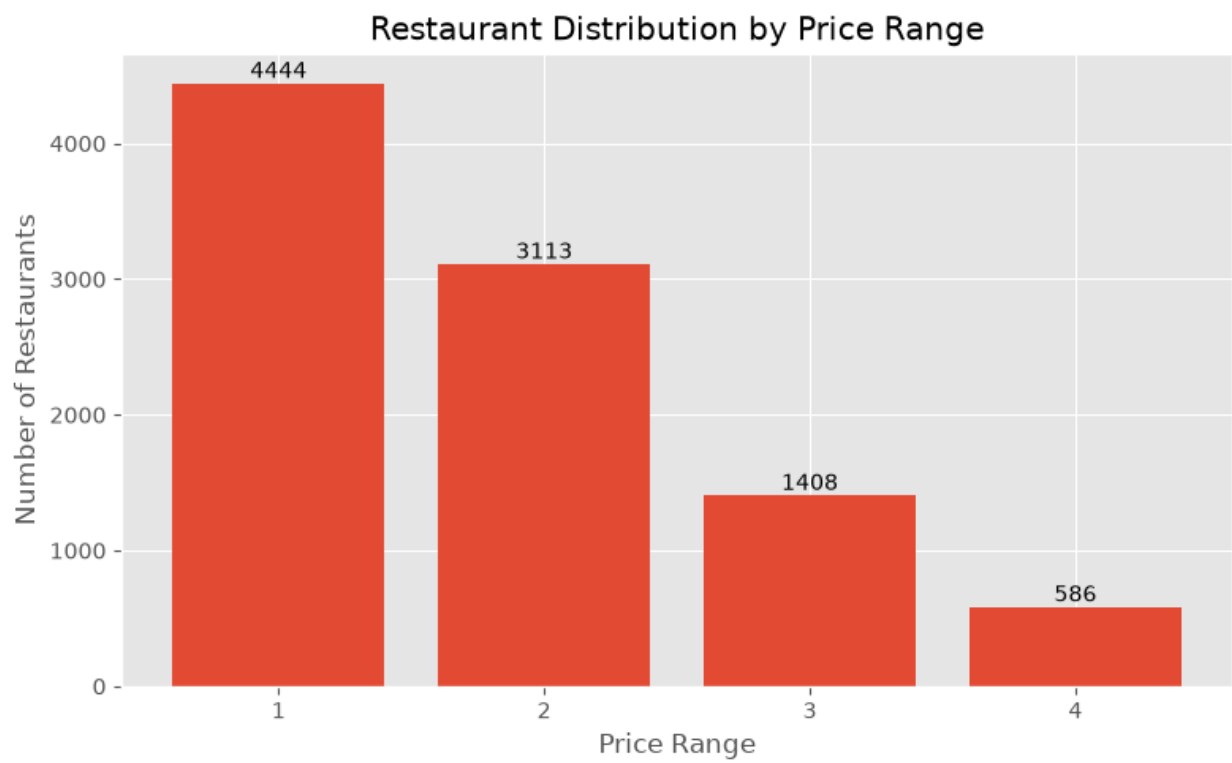
Top Cuisines



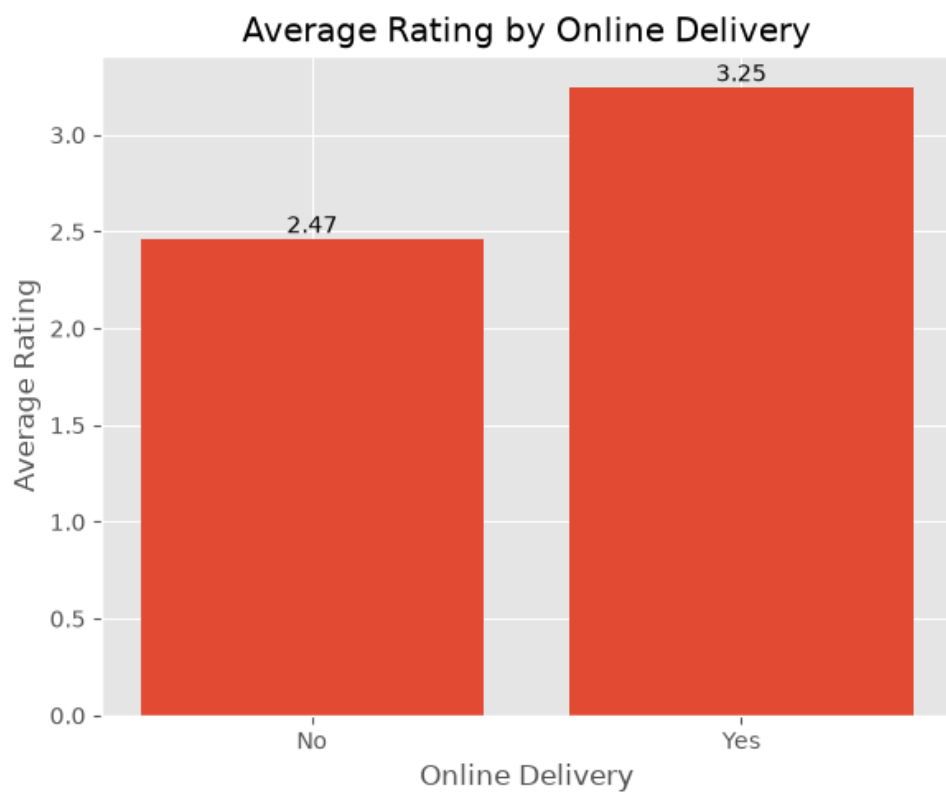
Top Cities by Restaurant Count



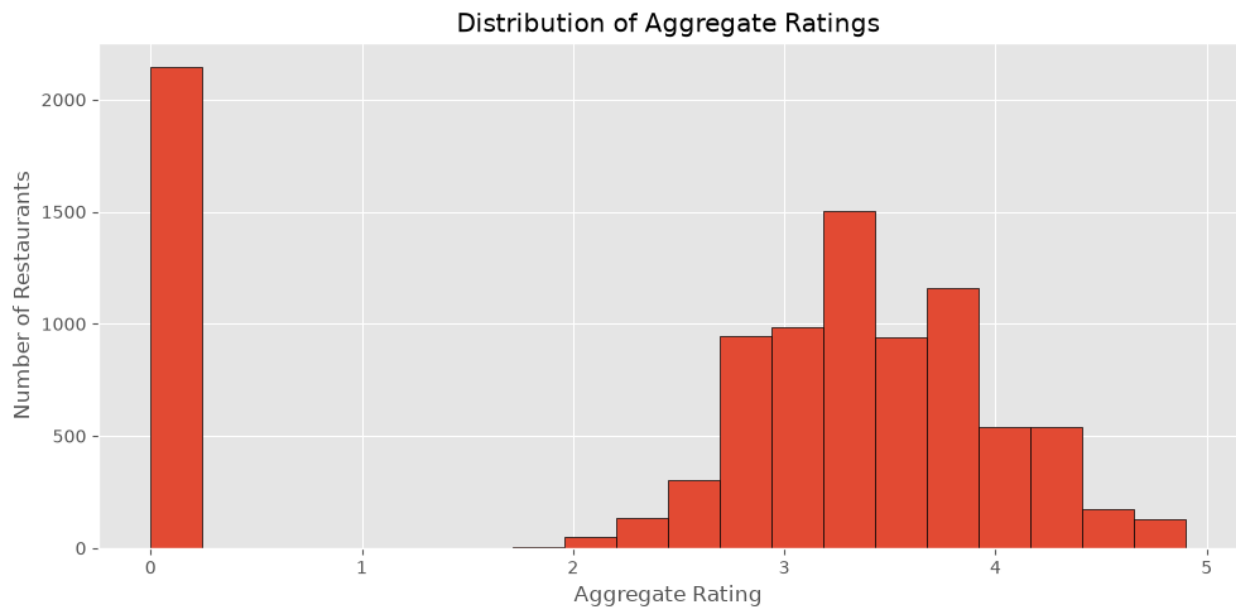
Price Range Distribution



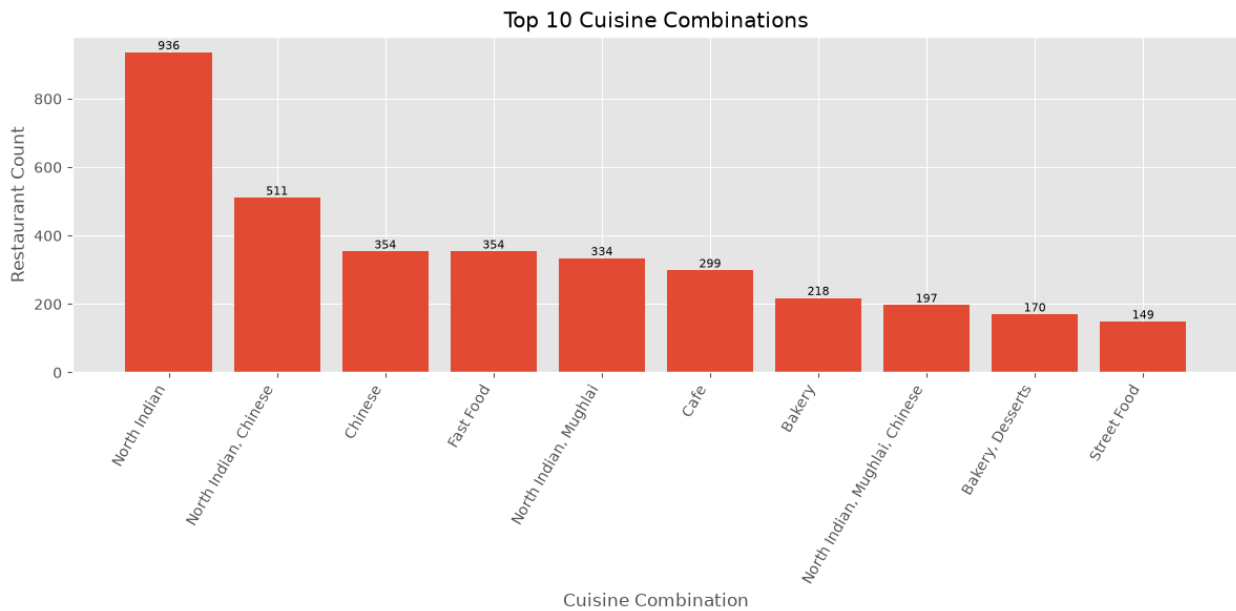
Average Rating by Online Delivery



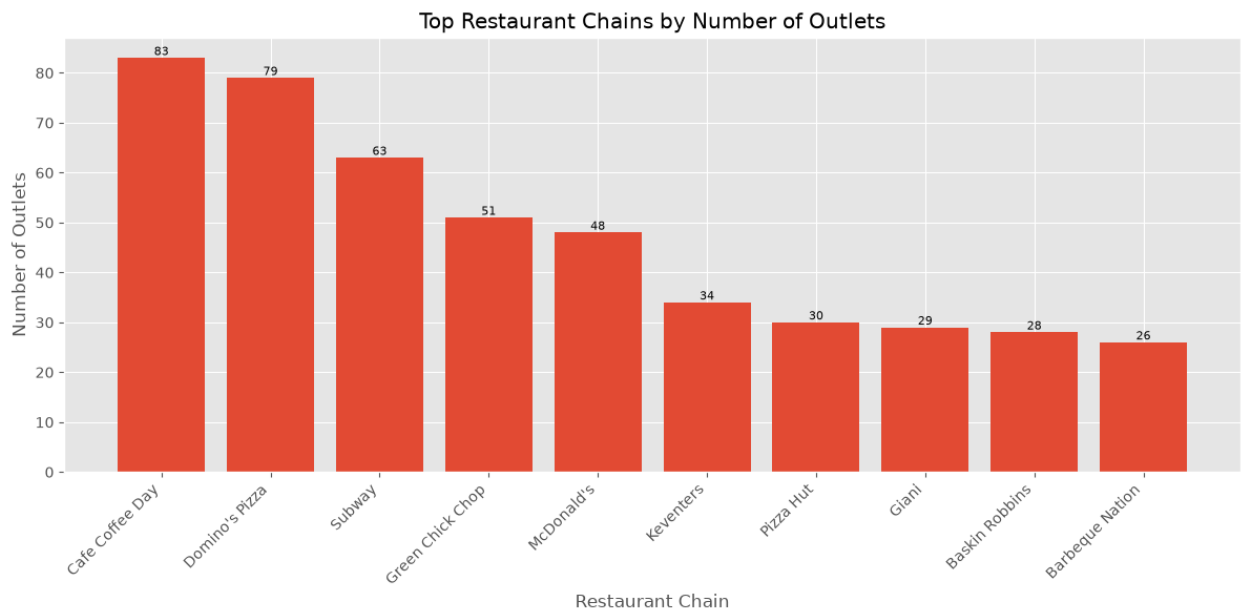
Restaurant Rating Distribution



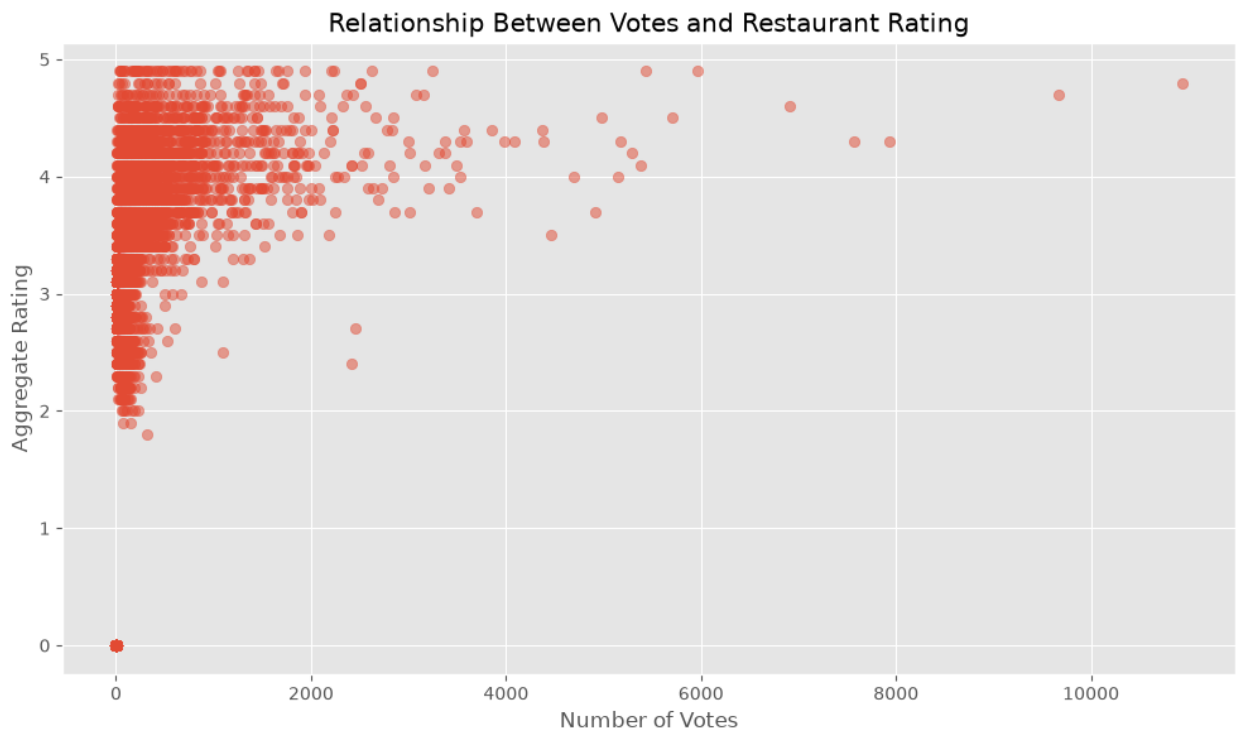
Common Cuisine Combinations



Top Restaurant Chains



Votes vs Restaurant Rating



Service Availability by Price Range



7. Conclusion

The completed analysis demonstrates an end-to-end exploratory data analysis workflow using Python. The results reveal strong concentration of restaurants in a few cities, a large share of restaurants in lower price ranges, meaningful differences in average ratings between restaurants with and without online delivery, and a moderate positive relationship between votes and aggregate rating. Table-booking availability increases substantially across the higher price ranges, whereas online delivery follows a non-monotonic pattern.

8. Project Deliverables

The project contains 11 Jupyter notebooks covering all three internship levels, a requirements file, README documentation, selected visualization assets, an interactive geographic map, and this final report.